

Week 8 Capstone Technical Report

This report documents a reproducible end-to-end analysis of sales, customer churn, and property pricing data. It is written for reviewers who need both the analytical method and the business meaning of the results.

1. Business Context and Problem Definition

The business needs a joined view of performance: sales teams need to understand product and regional revenue, customer teams need to understand which segments show higher churn, and property teams need evidence for pricing decisions. The datasets are separate samples, so the analysis answers each business question independently rather than manufacturing a relationship between unrelated records.

Success is defined as producing traceable metrics, clear visual evidence, practical recommendations, and a repeatable artifact-generation script. Key monitoring measures are total sales, product and regional revenue, churn rate by contract, average property price by location, and pricing relationships with physical property characteristics.

2. Dataset Inventory

The house-price dataset contains 300 properties and 8 columns. It includes area, bedrooms, bathrooms, age, location, property type, and price. The customer dataset contains 500 records and 9 columns covering tenure, charges, contract, payment method, senior-citizen status, and churn. The sales dataset contains 100 transactions and 7 original columns, with dates from 2024-01-01 through 2024-04-09 and one derived Month field.

All three files were loaded from the project data directory. Each dataset had zero missing values and zero duplicate records. The sales Date column was converted to datetime and used to derive monthly totals. Cleaned copies were written to cleaned_data without overwriting the supplied source files.

3. Data Preparation and Quality Checks

The preparation process validates file presence, reads each CSV with Pandas, checks for missing values and duplicates, converts the sales date, and saves reproducible cleaned copies. Numeric and categorical fields were retained because the supplied data passed the quality checks. April sales are treated as a partial period because the final sales date is April 9.

The datasets do not have a shared customer or property key. Customer IDs in the churn data are not present in the sales data, and property records have no sales key. Therefore, joins were intentionally avoided. This prevents unsupported customer-level sales conclusions and keeps each business analysis valid for its own source.

4. Sales Performance Analysis

Sales revenue totals 12,365,048 across 478 units. Laptop is the leading product, and North is the leading region. The monthly sequence is 2024-01: 4,120,524, 2024-02: 2,656,050, 2024-03: 4,485,006, 2024-04: 1,103,468. March is the strongest complete month; April cannot be compared as a full month.

Product	Total Sales
Laptop	3,889,210
Tablet	2,884,340
Phone	2,859,394
Headphones	1,384,033
Monitor	1,348,071

Region	Total Sales
North	3,983,635
South	3,737,852
East	2,519,639
West	2,123,922

Sales interpretation

Revenue is concentrated in a small number of product and regional leaders. That concentration creates an opportunity to protect availability and execution in leading segments, while using lower-performing segments as a focused improvement agenda. The analysis does not claim that product or region causes revenue differences; it identifies where management attention is most likely to have value.

5. Customer Churn and Retention Analysis

The observed churn rate is 10.60%. Contract-level rates are Month-to-month: 20.59%, Two year: 6.94%, One year: 4.30%. The gap between month-to-month and one-year contracts is a clear segmentation signal for retention planning. It should be monitored over time and tested with controlled campaigns before claiming causal impact.

Contract	Churn Rate
Month-to-month	20.59%
Two year	6.94%
One year	4.30%

Retention recommendation

Create an early-tenure monitoring list for month-to-month customers, provide clear upgrade value, and measure contact, conversion, and subsequent churn. Keep payment and billing attributes in the monitoring workflow, but avoid treating group differences as proof of customer behavior causes.

6. Property Pricing Analysis

The property dataset contains 300 observations. Mean price is 24,883,658, with a range from 3,695,000 to 58,700,000. City Center has the highest mean price at 33,149,794. The Pearson correlation between Area and Price is 0.796, indicating a strong positive linear association in this sample.

Location	Mean Price
City Center	33,149,794
Suburb	25,108,071
Rural	16,461,429

Pricing interpretation

Area is a useful starting variable for pricing review, while location adds a clear market-segmentation perspective. The project recommends using both features in a pricing review model, then validating predictions against future transactions. A correlation is informative but does not prove that increasing area alone causes a specific price increase.

7. Analysis Techniques and Statistical Reasoning

The project uses exploratory data analysis, descriptive statistics, groupby aggregation, monthly trend analysis, segmentation by contract, correlation analysis, and linear regression. These techniques were selected because the datasets contain numeric measures, categorical groupings, a time field in sales, and repeated business dimensions suitable for comparisons.

The analysis deliberately avoids artificial joins. It also distinguishes observed patterns from causal conclusions. This is important for a portfolio project: reproducibility and honest limitations are more valuable than a complicated but unsupported model.

8. Visualization Evidence

Five visualizations were generated from the actual values. `sales_analysis.png` combines the monthly sales trend and product performance. `churn_analysis.png` compares contract churn and monthly charges by churn status. `property_price_analysis.png` combines area-price observations and mean prices by location. `correlation_heatmap.png` shows relationships among property variables and price. `additional_visualization.png` shows regional sales performance.

Each chart has a focused business purpose: identify trends, compare groups, locate concentration, inspect relationships, or support pricing decisions. The chart files are saved under visualizations and can be inserted into a portfolio or presentation.

9. Business Insights

Sales: Laptop is the top product and North is the top region. This suggests protecting supply and execution in these leading segments while testing targeted improvement actions in lower-performing segments. Monthly sales peak in March, but April is partial and should not be interpreted as a complete-month decline.

Retention: Month-to-month customers have the highest observed churn rate. This supports early-retention outreach, clearer upgrade communication, and recurring segment monitoring. The data supports prioritization, not a claim that contract type alone causes churn.

Property: City Center has the highest average price, and Area has a strong positive association with Price. Pricing teams can use area and location together for market review, while validating the approach with new listings and completed transactions.

10. Actionable Recommendations

Recommendation 1 - High priority: establish a weekly churn dashboard focused on month-to-month customers and early-tenure signals. Recommendation 2 - Medium priority: align inventory and regional sales planning with Laptop and North-region performance. Recommendation 3 - Medium priority: introduce an area-and-location pricing review using comparable properties and track prediction error.

11. Implementation Plan

Action	Priority	Timeline	Success metric
Churn monitoring and outreach	High	0-30 days	Churn rate and response rate
Product and region planning	Medium	31-60 days	Product/region revenue
Property pricing review	Medium	31-90 days	Pricing error and conversion
Governance review	High	Monthly	Data quality and refresh completion

12. Limitations and Risk Controls

The sales sample contains 100 transactions and only covers part of April. The customer, sales, and property datasets are independent and cannot be joined. The analysis is observational, so correlations and group differences are not causal proof. Future work should add more time periods, establish reliable business keys, and test recommendations through controlled measurement.

13. Reproducibility and Quality Assurance

capstone_analysis.py loads the supplied files, validates missing values and duplicates, saves cleaned copies, calculates metrics, generates charts, and writes the PDF and PowerPoint deliverables. The final run completed successfully with 300 property rows, 500 customer rows, and 100 sales rows. Source CSVs were not overwritten.

14. Conclusion

This capstone turns three real datasets into a practical business portfolio. It identifies sales concentration, a retention priority, and a property pricing signal, then connects each finding to an action, owner-facing metric, and implementation timeline. The project is ready for review and presentation after manual screenshots are added.